Skin Cancer Classification Project Report

Comparative Analysis of Traditional Machine Learning and Deep Neural Networks

Dataset: HAM10000 | **Focus:** Multiclass Dermatoscopic Classification

1. Introduction

In this project, the **HAM10000** ("Human Against Machine with 10000 training images") skin cancer dataset was utilized to classify various types of skin lesions using both traditional machine learning algorithms and deep neural network architectures. Given the critical nature of automated dermatological diagnosis, the primary objective was to rigorously compare baseline statistical models against a multi-layer perceptron to identify the optimal pipeline for performance and deployment readiness.

2. Dataset Description

The dataset consists of thousands of high-quality dermatoscopic images paired with corresponding clinical metadata. The target variable is `dx`, representing the specific category of skin lesion.

A defining characteristic of this dataset is its significant class imbalance, reflecting real-world clinical prevalence where certain benign categories outnumber rarer malignant cases. Because standard accuracy metrics fail to reflect true performance under severe skewness, the **F1-score** (macro-averaged) was selected as the primary evaluation metric to ensure precision and recall are harmonized across all diagnostic categories.

3. Models Used & Baseline Evaluation

Three traditional machine learning models were established as baseline classifiers, leveraging distinct mathematical approaches to decision boundaries:

- **Logistic Regression:** A foundational linear model optimized with cross-entropy loss.
- **Random Forest:** An ensemble technique utilizing bagging and randomized feature splits to build uncorrelated decision trees.
- **K-Nearest Neighbors (KNN):** A non-parametric instance-based classifier relying on Euclidean distance metrics.

4. Neural Network Architecture

To capture highly non-linear feature interactions, a deep neural network was designed using the Keras framework. The architecture utilizes sequential **Dense layers** heavily reliant on **ReLU (Rectified Linear Unit)** activation functions to propagate gradients effectively.

To counter the inherent risk of overfitting associated with complex architectures, a structured **Dropout layer** was integrated to randomly deactivate nodes during training passes. The final output layer leverages a **Softmax activation function**, yielding a normalized probability distribution over all target classes for explicit multi-class inference. The model reached convergence over a robust training cycle of 10 epochs.

5. Results & Comparative Analysis

The empirical evaluations demonstrate a clear hierarchical progression in predictive capabilities, summarized in the comparative matrix below:

Classification Model	Core Framework / Paradigm	Achieved F1-Score	Performance Status
K-Nearest Neighbors (KNN)	Instance-based / Distance	0.75	Baseline Linearity Limit
Logistic Regression	Generalized Linear Model	0.79	Moderate Predictability
Random Forest	Ensemble Learning (Bagging)	0.84	Best Traditional Performer
Neural Network (MLP)	**Deep Learning / Keras Dense**	**0.85**	**Optimal Overall Champion**

The **Neural Network** achieved the top overall performance with an F1-score of approximately **0.85**. While it closely contested with the Random Forest model, the Deep Learning approach exhibited vastly superior precision and proved significantly more capable of learning intricate structural patterns native to the tabular representation of the dataset, albeit at the expense of computational training duration.

6. Key Project Insights & Learnings

This project provided valuable end-to-end exposure to the data science life cycle. Key competencies mastered include handling real-world class imbalances through strategic metric alignment, configuring standard data preprocessing pipelines, optimizing traditional ensemble classifiers, and crafting custom multi-layer deep networks via Keras. The empirical results solidified the understanding that while traditional models like Random Forest offer exceptional performance with minimal compute overhead, neural architectures provide the necessary representation depth required for complex medical classification challenges.